

SC3021 Exercise: Data Science Ecosystems

Discussed in Week 4 Tutorial

Data Science Problem Setting

In efforts to automatically generate reports for various government agencies, your company is contracted with developing an image recognition system that detects and transcribes handwritten text from scanned documents to digital form. The system shall be deployed in government archives to digitize historical records efficiently and effectively.

- Data engineering: list one example for required functionality for each data preparation step (i.e., profiling, structuring, enriching, cleaning)
- Data engineering: What data should be managed by a data management system? Explain how the required system(s) may differ from the relational data management systems introduced in class?
- Which type(s) of data analytics need to be considered for this application? Briefly explain your answer.
- Data understanding: Provide an example of an explanation that can support data understanding. Which users benefiting from it do you have in mind? Why is the proposed explanation useful to them?
- Data understanding: Provide an example of a visualization that can support data understanding. Which users benefiting from it do you have in mind? Why is the proposed visualization useful to them?
- Data protection: Give at least one example of a specific data security issue that you need to take into account during system development.
- Data protection: Give at least one example of a specific privacy issue that you need to take into account during system development.
- Data ethics: Give at least one example of a specific data ethics issue that may arise. Is it related to data, algorithms, or practices?

DATA SCIENCE: A REDUCTIONIST VIEW

```
1 import pandas as pd
2 import sklearn
3
4 ... # do something|
```

DATA SCIENCE: A COMPLETE VIEW (BOSS')

Data engineering

- Data preparation
- Data management

Data analytics

- Data exploration (data mining)
- Models and algorithms (machine learning)

Data understanding

- Explanation
- Visualization

Data protection

- Security
- Privacy

Data ethics

- Impact on individuals, organizations, and society
- Bias in data
- Algorithmic bias
- Regulatory issues

DEFINE THE PROBLEM

In efforts to automatically generate reports for various government agencies, your company is contracted with developing an image recognition system that detects and transcribes handwritten text from scanned documents to digital form. The system shall be deployed in government archives to digitize historical records efficiently and effectively.



What features do we want the system to have?

Hint: Be creative, be a boss

FEATURES

- A web interface to upload document scans and get transcripts in PDF, DOCX, etc...
- A search engine to search through the scanned documents
- Password-protection to restrict access to sensitive documents

DATA SCIENCE BOSS

Data engineering

- Data preparation
- Data management

Data analytics

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TYPES OF DATA ANALYTICS



What kind of analytics is this?

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KINDS OF ANALYTICS

Predictive: get the text
from the image

Descriptive: get a
description of the
handwritten document

WHICH DATA DO WE NEED ?????

In efforts to automatically generate reports for various government agencies, your company is contracted with developing an image recognition system that detects and transcribes handwritten text from scanned documents to digital form. The system shall be deployed in government archives to digitize historical records efficiently and effectively.

WHICH DATA DO WE NEED?



The company probably already has an image-to-text model, but might not be trained on administrative languages or notations; or old archives



The government provides the company with a representative sample of handwritten documents, with (possibly) their transcriptions



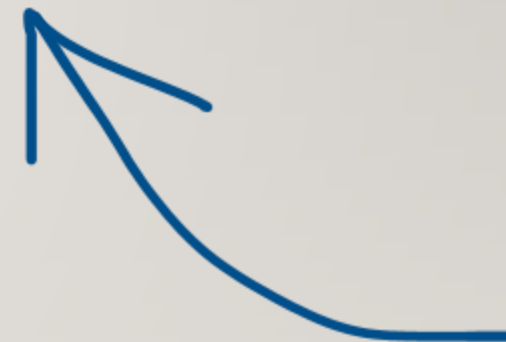
What to do?

Data engineering	Data analytics	Data understanding	Data protection	Data ethics
- Data preparation - Data management	- Data exploration (data mining) - Models and algorithms (machine learning)	- Explanation - Visualization	- Security - Privacy	- Impact on individuals, organizations, and society - Bias in data - Algorithmic bias - Regulatory issues

- Data engineering: list one example for required functionality for each data preparation step (i.e., profiling, structuring, enriching, cleaning)



The government provides the company with a representative sample of handwritten documents, with (possibly) their transcriptions



DATA PREPARATION



Profiling: investigate document types, languages; identify issues such as faded pages or stained papers



Enriching: convert images to text with existing tools



Cleaning: use spell checker on resulting texts; improve images through filters/contrast improvements

- Data engineering: What data should be managed by a data management system? Explain how the required system(s) may differ from the relational data management systems introduced in class?

Name	Kind
 civil_code_1992_page_1.jpg	JPEG image
 civil_code_1992_page_1.txt	text
 government_report_1995.jpg	JPEG image
 government_report_1995.txt	text

Relational data management systems = tables;

Here we need some other systems, such as e.g. a file system

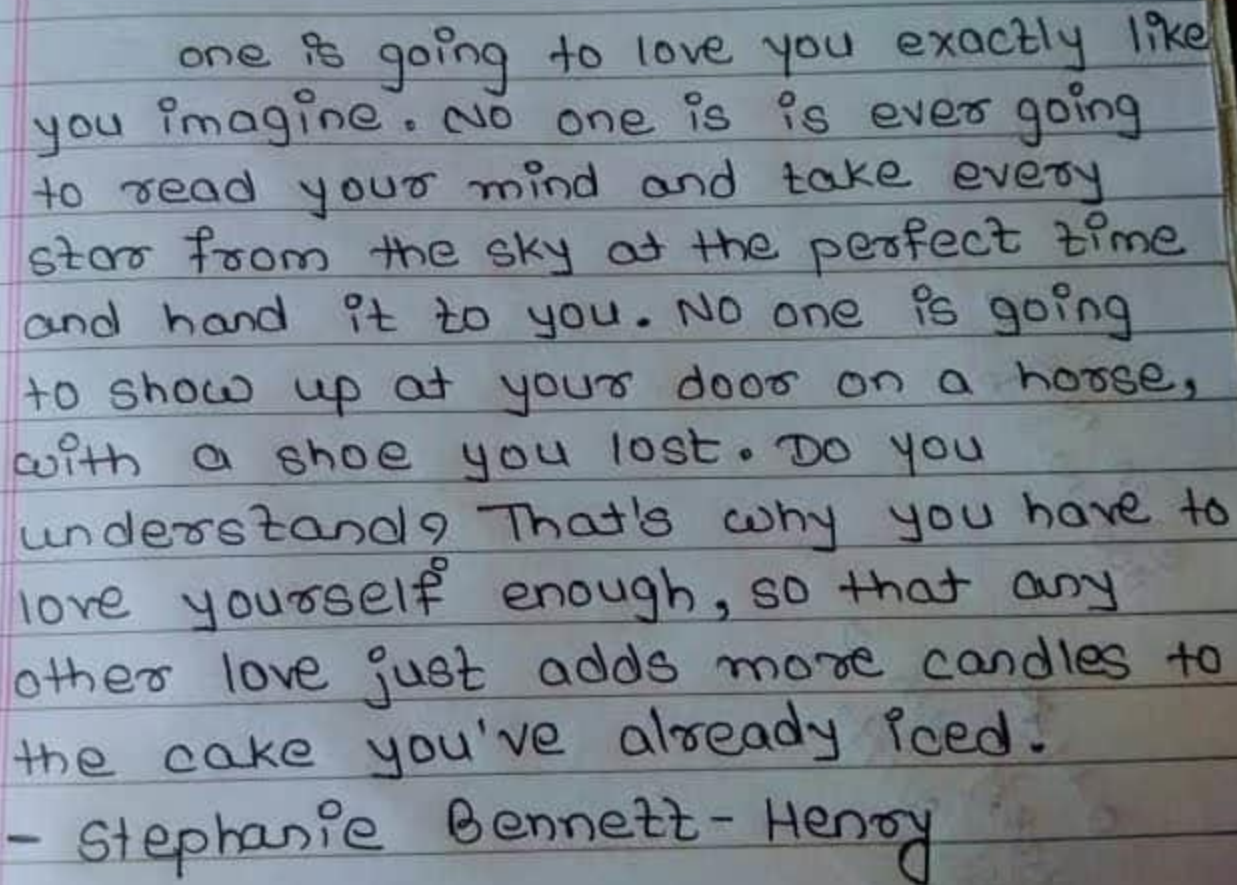
Name	Kind
 civil_code_1992_page_1.jpg	JPEG image
 civil_code_1992_page_1.txt	text
 government_report_1995.jpg	JPEG image
 government_report_1995.txt	text

How to visualize? Be creative!

- Data understanding: Provide an example of a visualization that can support data understanding. Which users benefiting from it do you have in mind? Why is the proposed visualization useful to them?

CONVERT HANDWRITING TO TEXT

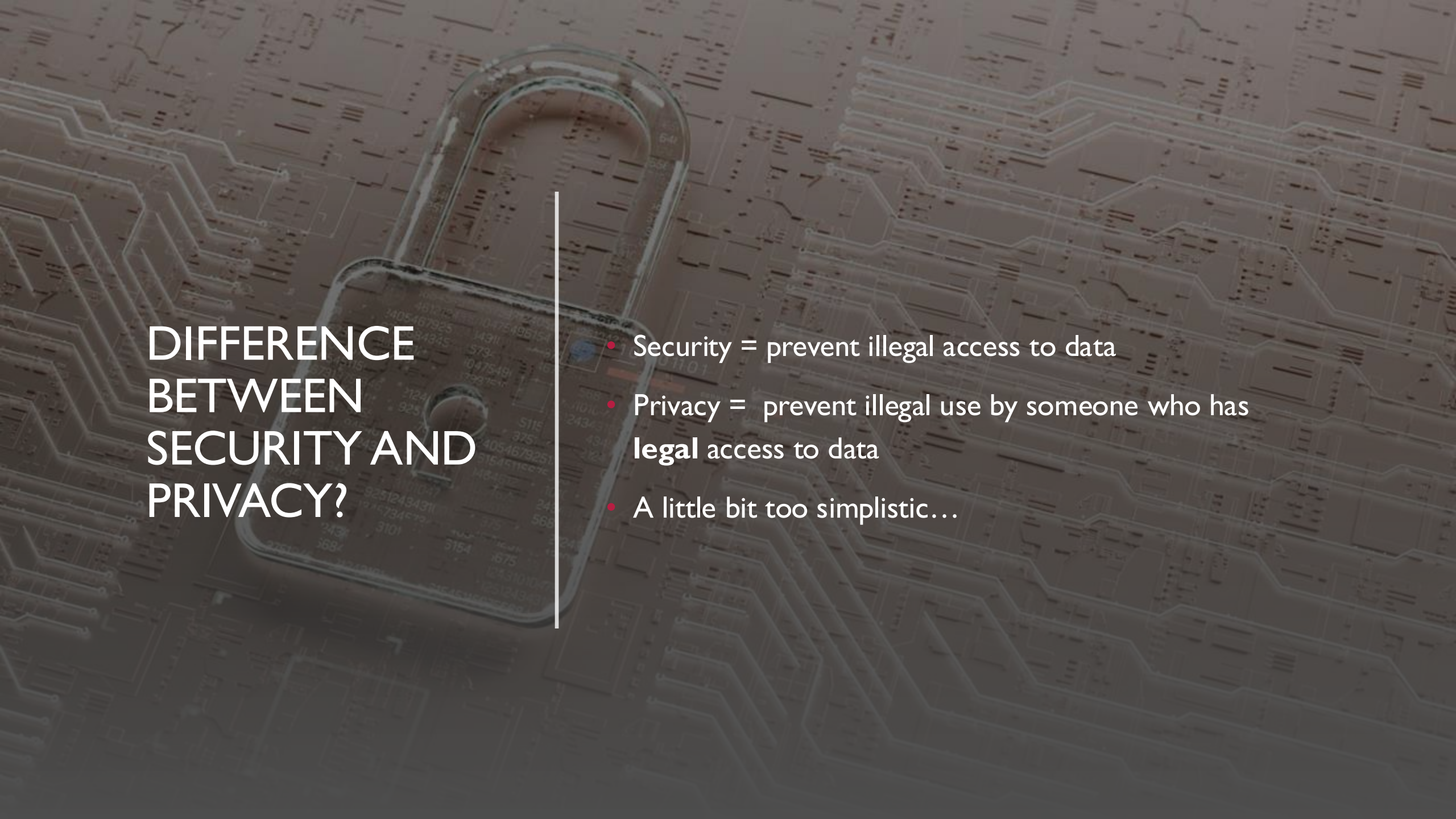
Before

A photograph of a piece of lined paper with handwritten text in cursive. The text is somewhat blurry and has some corrections. A large red arrow points from this section towards the 'After' section.

one is going to love you exactly like
you imagine. No one is is ever going
to read your mind and take every
star from the sky at the perfect time
and hand it to you. No one is going
to show up at your door on a horse,
with a shoe you lost. Do you
understand? That's why you have to
love yourself enough, so that any
other love just adds more candles to
the cake you've already iced.
- Stephanie Bennett - Hendy

After

one is going to love you exactly like you imagine.
No one is ever going to read your mind and take
every story from the sky at the perfect zyme and
hand it to you. No one is going to show up at
one is going to love you exactly like you imagine.
hand it to you. No one is going to show up at
one is going to love you exactly like you imagine.
No one is ever going to read your mind and take
every story from the sky at the perfect zyme and
hand it to you. No one is going to show up at
your door on a horse, with a shoe you lost. Do
you understand That's why you have to love
yourself enough, so that any other love just adds
Stephanie Bennett - Hendy



DIFFERENCE BETWEEN SECURITY AND PRIVACY?

- Security = prevent illegal access to data
- Privacy = prevent illegal use by someone who has **legal** access to data
- A little bit too simplistic...

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SECURITY CONCERNS?

PRIVACY CONCERNS?



SECURITY AND PRIVACY

- Security: government data can be sensitive, needing appropriate access control
- Privacy: make sure private data doesn't enter into training procedure

- Data understanding: Provide an example of an explanation that can support data understanding. Which users benefiting from it do you have in mind? Why is the proposed explanation useful to them?

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Example: How to measure transcription quality?

- Compare text to see the differences
- Test by document types
- Get, e.g. proportion of wrong/missing transcription per doc types

1 The quick brown fox jumps over the lazy dog.

1 The yellow fox jumps over the smart handsome dog.

Edit Distance Example

enjoyalgorithms.com

